

# Solving Linear Systems by Substitution

## Answer Key

### Algebra 1

#### Quick Answers

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1.  $x = 3, y = 6$

2.  $x = 3, y = -1$

3.  $x = 4, y = 3$

4.  $x = 16, y = 5$

5.  $x = 3, y = 4$

6.  $x = 7, y = 5$

7.  $x = \frac{7}{5}, y = \frac{11}{5}$

8.  $x = 5, y = 2$

9. —

10.  $x = 2, y = 2$

11.  $x = 6, y = 17$

12.  $x = 8, y = 4$ 

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#### Curriculum

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**Level:** Algebra 1

HSA-REI.C.6 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

*Difficulty 1–5 of 5 across 12 tagged checks.*

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**1.**  $y = 2x, \quad 3x + y = 15$

**Step 1.** The first equation is already solved for  $y$ , so replace  $y$  in the second:  $3x + 2x = 15$ .**Step 2.**  $5x = 15$ , so  $x = 3$ .**Step 3.** Back-substitute:  $y = 2(3) = 6$ . Check in  $3x + y$ :  $9 + 6 = 15$  ✓

(3, 6)

**2.**  $y = x - 4, \quad 2x + 3y = 3$

**Step 1.** Substitute  $x - 4$  for  $y$ :  $2x + 3(x - 4) = 3$ .**Step 2.** Distribute and collect:  $2x + 3x - 12 = 3$ , so  $5x = 15$  and  $x = 3$ .**Step 3.** Back-substitute:  $y = 3 - 4 = -1$ . Check in  $2x + 3y$ :  $6 - 3 = 3$  ✓

(3, -1)

**3.**  $2x + y = 11, \quad 3x + 2y = 18$

**Step 1.** Nothing is isolated yet. The cheapest isolation is  $y$  in the first equation (its coefficient is 1):  $y = 11 - 2x$ .**Step 2.** Substitute into the second:  $3x + 2(11 - 2x) = 18$ , so  $3x + 22 - 4x = 18$  and  $-x = -4$ , giving  $x = 4$ .**Step 3.** Back-substitute:  $y = 11 - 8 = 3$ . Check in  $3x + 2y$ :  $12 + 6 = 18$  ✓

(4, 3)

**4.**  $x = 3y + 1, \quad 2x - 5y = 7$

**Step 1.** Here  $x$  is the isolated variable, so substitute for  $x$ :  $2(3y + 1) - 5y = 7$ .**Step 2.**  $6y + 2 - 5y = 7$ , so  $y + 2 = 7$  and  $y = 5$ .**Step 3.** Back-substitute:  $x = 3(5) + 1 = 16$ . Check in  $2x - 5y$ :  $32 - 25 = 7$  ✓

(16, 5)

5.  $x + y = 7$ ,  $3x - y = 5$

**Step 1.** Isolate  $y$  in the first equation:  $y = 7 - x$ .

**Step 2.** Substitute:  $3x - (7 - x) = 5$ . The subtraction applies to the whole expression, so  $3x - 7 + x = 5$ , giving  $4x = 12$  and  $x = 3$ .

**Step 3.** Back-substitute:  $y = 7 - 3 = 4$ . Check in  $3x - y$ :  $9 - 4 = 5$  ✓

$$(3, 4)$$

6. Tickets:  $x$  adult at 8 dollars,  $y$  child at 5 dollars.

**Step 1.** Counting tickets gives  $x + y = 12$ ; counting dollars gives  $8x + 5y = 81$ .

**Step 2.** Isolate in the count equation:  $y = 12 - x$ , then substitute:  $8x + 5(12 - x) = 81$ .

**Step 3.**  $8x + 60 - 5x = 81$ , so  $3x = 21$  and  $x = 7$ ; then  $y = 12 - 7 = 5$ . Check the money:  $56 + 25 = 81$  ✓

The family bought 7 adult tickets and 5 child tickets.

$$(7, 5)$$

7.  $y = 3x - 2$ ,  $2x + y = 5$

**Step 1.** Substitute for  $y$ :  $2x + (3x - 2) = 5$ .

**Step 2.**  $5x - 2 = 5$ , so  $5x = 7$  and  $x = \frac{7}{5}$ . A fraction is a perfectly good solution — do not round it.

**Step 3.** Back-substitute:  $y = 3 \cdot \frac{7}{5} - 2 = \frac{21}{5} - \frac{10}{5} = \frac{11}{5}$ . Check in  $2x + y$ :  $\frac{14}{5} + \frac{11}{5} = \frac{25}{5} = 5$  ✓

$$\left(\frac{7}{5}, \frac{11}{5}\right)$$

8.  $x - 3y = -1$ ,  $2x + y = 12$

**Step 1.** Isolate  $x$  in the first equation:  $x = 3y - 1$ .

**Step 2.** Substitute:  $2(3y - 1) + y = 12$ , so  $6y - 2 + y = 12$  and  $7y = 14$ , giving  $y = 2$ .

**Step 3.** Back-substitute:  $x = 3(2) - 1 = 5$ . Check in  $2x + y$ :  $10 + 2 = 12$  ✓

$$(5, 2)$$

9.  $y = 2x + 3$ ,  $4x - 2y = 1$  — **no solution**

**Step 1.** Substitute for  $y$ :  $4x - 2(2x + 3) = 1$ .

**Step 2.**  $4x - 4x - 6 = 1$ , so  $-6 = 1$ . Both variables vanished and what is left is *false*.

**Step 3.** A false statement means no  $(x, y)$  can satisfy both equations: the system has **no solution**. Rewriting the second equation as  $y = 2x - \frac{1}{2}$  shows why — same slope 2, different intercepts, so the lines are parallel and never meet.

*Answer to accept:* “No solution; the lines are parallel” with the  $-6 = 1$  collapse shown as evidence.

10.  $y = 3x - 4$ ,  $6x - 2y = 8$  — **infinitely many**

**Step 1.** Substitute for  $y$ :  $6x - 2(3x - 4) = 8$ .

**Step 2.**  $6x - 6x + 8 = 8$ , so  $8 = 8$ . The variables vanish again, but this time the statement is *true*.

**Step 3.** A true statement means every point of the line satisfies both equations: the two equations describe the same line, so there are infinitely many solutions — all pairs with  $y = 3x - 4$ . Taking  $x = 2$  gives  $y = 3(2) - 4 = 2$ , one specific solution.

$$(2, 2)$$

**11.** Rectangle: width  $x$ , length  $y$ , perimeter 46 m.

**Step 1.** “Length is 5 more than twice the width” gives  $y = 2x + 5$ ; the perimeter gives  $2y + 2x = 46$ .

**Step 2.** The first is already isolated, so substitute:  $2(2x + 5) + 2x = 46$ , so  $4x + 10 + 2x = 46$  and  $6x = 36$ , giving  $x = 6$ .

**Step 3.** Back-substitute:  $y = 2(6) + 5 = 17$ . Check the perimeter:  $2(17) + 2(6) = 34 + 12 = 46$  ✓

The rectangle is 6 m wide and 17 m long.

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| $(6, 17)$ |
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**12.** Mixture:  $x$  litres of 20% acid,  $y$  litres of 50% acid.

**Step 1.** The volumes must total 12 L, so the missing equation is  $x + y = 12$ ; the acid balance is given as  $0.2x + 0.5y = 3.6$ .

**Step 2.** Isolate in the volume equation:  $y = 12 - x$ , then substitute:  $0.2x + 0.5(12 - x) = 3.6$ .

**Step 3.**  $0.2x + 6 - 0.5x = 3.6$ , so  $-0.3x = -2.4$  and  $x = 8$ ; then  $y = 12 - 8 = 4$ . Check the acid:  $0.2(8) + 0.5(4) = 1.6 + 2 = 3.6$  ✓

The chemist uses 8 L of the 20% solution and 4 L of the 50% solution.

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| $(8, 4)$ |
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